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In [38]: from sklearn.datasets import load_iris
```

```
In [39]: iris=load_iris()
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```
In [41]: X=iris.data
```

```
In [42]: y=iris.target  
y
```

```
Out[42]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
                0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
                1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2,  
                2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,  
                2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2])
```

```
In [43]: from sklearn.model_selection import train_test_split  
X_train, X_test, y_train, y_test=train_test_split(X,y, test_size=0.2)
```

```
In [45]: X_train.shape
```

```
Out[45]: (120, 4)
```

```
In [46]: X_test.shape
```

```
Out[46]: (30, 4)
```

```
In [47]: from sklearn.tree import DecisionTreeClassifier
```

```
In [48]: clf=DecisionTreeClassifier()
```

```
In [49]: clf.fit(X_train,y_train)
```

```
Out[49]: DecisionTreeClassifier(class_weight=None, criterion='gini', max_depth=None,  
                                max_features=None, max_leaf_nodes=None,  
                                min_impurity_decrease=0.0, min_impurity_split=None,  
                                min_samples_leaf=1, min_samples_split=2,  
                                min_weight_fraction_leaf=0.0, presort=False,  
                                random_state=None, splitter='best')
```

```
In [50]: y_pred=clf.predict(X_test)
```

```
In [51]: from sklearn.metrics import accuracy_score  
accuracy_score(y_test,y_pred)
```

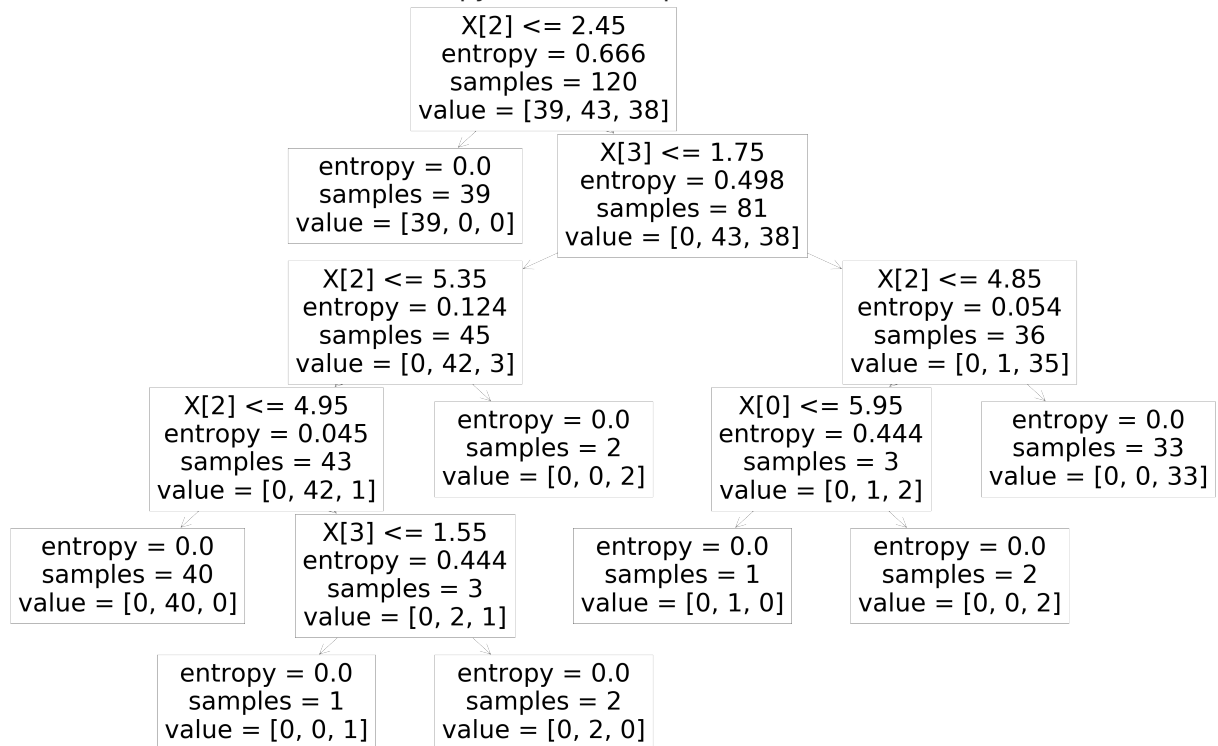
```
Out[51]: 0.9666666666666667
```

```
In [52]: from sklearn.tree import plot_tree
```

```
In [53]: from matplotlib.pylab import rcParams
rcParams['figure.figsize'] = 80,50

plot_tree(clf)
```

```
Out[53]: [Text(1984.0, 2491.5, 'X[2] <= 2.45\nentropy = 0.666\nsamples = 120\nvalue = [39, 43, 38]'),
Text(1488.0, 2038.5, 'entropy = 0.0\nsamples = 39\nvalue = [39, 0, 0]'),
Text(2480.0, 2038.5, 'X[3] <= 1.75\nentropy = 0.498\nsamples = 81\nvalue = [0, 43, 38]'),
Text(1488.0, 1585.5, 'X[2] <= 5.35\nentropy = 0.124\nsamples = 45\nvalue = [0, 42, 3]'),
Text(992.0, 1132.5, 'X[2] <= 4.95\nentropy = 0.045\nsamples = 43\nvalue = [0, 42, 1]'),
Text(496.0, 679.5, 'entropy = 0.0\nsamples = 40\nvalue = [0, 40, 0]'),
Text(1488.0, 679.5, 'X[3] <= 1.55\nentropy = 0.444\nsamples = 3\nvalue = [0, 2, 1]'),
Text(992.0, 226.5, 'entropy = 0.0\nsamples = 1\nvalue = [0, 0, 1]'),
Text(1984.0, 226.5, 'entropy = 0.0\nsamples = 2\nvalue = [0, 2, 0]'),
Text(1984.0, 1132.5, 'entropy = 0.0\nsamples = 2\nvalue = [0, 0, 2]'),
Text(3472.0, 1585.5, 'X[2] <= 4.85\nentropy = 0.054\nsamples = 36\nvalue = [0, 1, 35]'),
Text(2976.0, 1132.5, 'X[0] <= 5.95\nentropy = 0.444\nsamples = 3\nvalue = [0, 1, 2]'),
Text(2480.0, 679.5, 'entropy = 0.0\nsamples = 1\nvalue = [0, 1, 0]'),
Text(3472.0, 679.5, 'entropy = 0.0\nsamples = 2\nvalue = [0, 0, 2]'),
Text(3968.0, 1132.5, 'entropy = 0.0\nsamples = 33\nvalue = [0, 0, 33]')]
```



```
In [54]: import numpy as np
import pandas as pd
```

```
In [55]: data=pd.read_csv('Social_Network_Ads.csv')
data.head()
```

```
Out[55]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [56]: data['Gender'].replace({'Male':0,'Female':1},inplace=True)
```

```
In [57]: X=data.iloc[:, 1:4].values
y=data.iloc[:, -1].values
```

```
In [58]: X.shape
```

```
Out[58]: (400, 3)
```

```
In [59]: y.shape
```

```
Out[59]: (400,)
```

```
In [64]: clf1=DecisionTreeClassifier(max_depth=3)
```

```
In [65]: clf1.fit(X,y)
```

```
Out[65]: DecisionTreeClassifier(class_weight=None, criterion='gini', max_depth=3,
                                max_features=None, max_leaf_nodes=None,
                                min_impurity_decrease=0.0, min_impurity_split=None,
                                min_samples_leaf=1, min_samples_split=2,
                                min_weight_fraction_leaf=0.0, presort=False,
                                random_state=None, splitter='best')
```

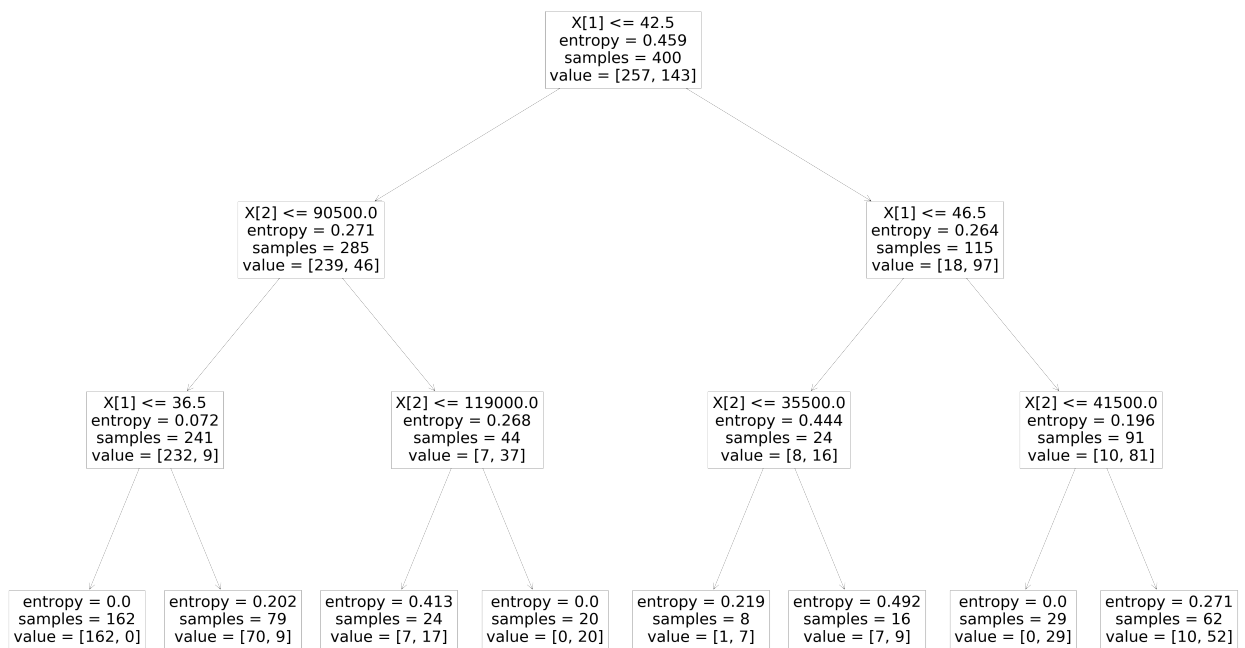
```
In [66]: rcParams['figure.figsize'] = 80,50

plot_tree(clf1)
```

```

Out[66]: [Text(2232.0, 2378.25, 'X[1] <= 42.5\nentropy = 0.459\nsamples = 400\nvalue =
[257, 143]'),
Text(1116.0, 1698.75, 'X[2] <= 90500.0\nentropy = 0.271\nsamples = 285\nvalue = [239, 46]'),
Text(558.0, 1019.25, 'X[1] <= 36.5\nentropy = 0.072\nsamples = 241\nvalue = [232, 9]'),
Text(279.0, 339.75, 'entropy = 0.0\nsamples = 162\nvalue = [162, 0]'),
Text(837.0, 339.75, 'entropy = 0.202\nsamples = 79\nvalue = [70, 9]'),
Text(1674.0, 1019.25, 'X[2] <= 119000.0\nentropy = 0.268\nsamples = 44\nvalue = [7, 37]'),
Text(1395.0, 339.75, 'entropy = 0.413\nsamples = 24\nvalue = [7, 17]'),
Text(1953.0, 339.75, 'entropy = 0.0\nsamples = 20\nvalue = [0, 20]'),
Text(3348.0, 1698.75, 'X[1] <= 46.5\nentropy = 0.264\nsamples = 115\nvalue = [18, 97]'),
Text(2790.0, 1019.25, 'X[2] <= 35500.0\nentropy = 0.444\nsamples = 24\nvalue = [8, 16]'),
Text(2511.0, 339.75, 'entropy = 0.219\nsamples = 8\nvalue = [1, 7]'),
Text(3069.0, 339.75, 'entropy = 0.492\nsamples = 16\nvalue = [7, 9]'),
Text(3906.0, 1019.25, 'X[2] <= 41500.0\nentropy = 0.196\nsamples = 91\nvalue = [10, 81]'),
Text(3627.0, 339.75, 'entropy = 0.0\nsamples = 29\nvalue = [0, 29]'),
Text(4185.0, 339.75, 'entropy = 0.271\nsamples = 62\nvalue = [10, 52]')]

```



In []: